Project Initialization and Planning Phase

Date	15 July 2026
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal outlines Rising Waters, a machine learning system that turns raw regional rainfall and cloud-cover readings into an instant, percentage-based flood-risk verdict. It addresses the lack of affordable, hyperlocal flood early-warning tools by pairing a trained XGBoost classifier with a lightweight, validated web interface that anyone can use from a browser.

Project Overview	
Objective	Predict regional flood risk instantly from five rainfall/cloud-cover readings using a trained ML classifier, delivered through a free, publicly accessible web app.
Scope	Covers collection and preprocessing of historical rainfall data, training and evaluating classification models, and building and deploying a Flask web application with validated input, real-time prediction, and clear result visualization.
Problem Statement	
Description	Many flood-prone regions lack affordable, localized early-warning tools; existing forecasts are broad, state-level, and delayed, leaving residents and local officials unprepared for fast-onset flood risk.
Impact	Solving this gives communities faster, low-cost access to flood-risk information, helping reduce property damage, improve preparedness, and speed up local response and evacuation decisions.
Proposed Solution	
Approach	Train an XGBoost classifier on historical rainfall data (cloud cover, annual rainfall, and seasonal rainfall blocks) using scikit-learn preprocessing (StandardScaler), then serve predictions through a validated Flask web form.
Key Features	- ML-based flood-risk classifier (XGBoost) trained on historical rainfall data - Real-time prediction returned via web form in under a second - Two-layer (client + server) input validation to prevent bad data reaching the model - Clean UI with an animated rain-gauge visual and circular probability gauges - Deployed live on Render, with source hosted on GitHub

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	CPU only (no GPU required); standard Render web service instance

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Memory	RAM specifications	512 MB – 1 GB
Storage	Disk space for data, models, and logs	< 100 MB (model artifacts + dataset)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, pandas, numpy, joblib
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Rainfall in India 1901–2015 (public), ~4,090 usable rows, xlsx/csv; flood dataset.xlsx (115 rows), csv